

Advanced Clustering in R: Fuzzy, Gaussian Mixture, and Hierarchical

Widening the Clustering Perspective

`clu_kmeans`, `clu_pam`, and `clu_dbscan` all share one assumption: every instance belongs to exactly one cluster, decided by geometric proximity in a feature matrix. This deck breaks that assumption in four different directions.

Soft Membership

An instance can partially belong to several clusters at once.

Probabilistic Structure

Clusters are modeled as overlapping statistical distributions, not hard regions.

Hierarchy

Clusters are nested at multiple resolutions, not fixed at one k .

Graphs, Not Points

The input is a network of relationships, not a table of coordinates.

Fuzzy Membership – cluster_cmeans

Fuzzy c-means clustering keeps membership degrees instead of only hard assignments — every instance gets a probability-like score for every cluster, not just one final label.

```
model <- cluster_cmeans(centers = 3, m = 2)
model <- fit(model, x)
clu <- cluster(model, x)
head(attr(clu, "membership"))
```

- **Fuzziness knob:** *m* controls fuzziness — higher values produce softer, more overlapping memberships.
- **Attached result:** the membership matrix rides along with the clustering result — useful when a hard yes/no call would hide genuinely ambiguous points.

Reference: Bezdek, J. C. (1981). Pattern Recognition with Fuzzy Objective Function Algorithms.

Probabilistic Mixtures – cluster_gmm

Gaussian Mixture Model clustering models the data as a mixture of overlapping Gaussian distributions rather than partitioning it by distance to a centroid or medoid.

```
model <- cluster_gmm(G = 3)
model <- fit(model, x)
clu <- cluster(model, x)
```

- **Component count:** G sets the number of mixture components (analogous to k in k-means).
- **Elliptical, overlapping:** clusters can be elliptical and overlapping — a natural fit when partitional methods' spherical assumption does not hold.

Reference: Fraley, C., and Raftery, A. E. (2002). Model-Based Clustering, Discriminant Analysis, and Density Estimation.

Hierarchical Structure – cluster_hclust

Agglomerative hierarchical clustering builds a full merge tree that can be inspected before or after cutting it into a fixed number of clusters — the dendrogram itself is part of the deliverable.

```
model <- cluster_hclust(  
  k = 3, method = "ward.D2")  
model <- fit(model, x)  
clu <- cluster(model, x)  
plot_dendrogram(model$hc,  
  title = "Hierarchical clustering")
```

- **Merge criterion:** method = "ward.D2" minimizes within-cluster variance at each merge step.
- **Cut after the fact:** the dendrogram lets the reader choose a different k later, without refitting.

Reference: Johnson, S. C. (1967). Hierarchical Clustering Schemes.

Graph Communities – cluster_louvain_graph

The one deliberate exception in the whole clustering family: the input is a graph, not a feature matrix, and the output is communities defined by modularity optimization, not geometric groups.

```
g <- igraph::sample_gnp(n = 20, p = 0.15)
model <- cluster_louvain_graph()
model <- fit(model, g)
clu <- cluster(model, g)
attr(clu, "modularity")
```

- **No feature matrix:** the model consumes an igraph object directly, not an iris[, 1:4]-style table.
- **Common mistake:** comparing this too literally against centroid-based clustering — they don't solve the same geometric problem.
- **Common mistake:** expecting a reference label like Species — many graph problems are fully unsupervised, with no ground truth.

Reference: Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). Fast Unfolding of Communities in Large Networks.

What to Observe

- ❏ • **The daltoolbox rhythm survives:** configure, fit, cluster, evaluate — even the graph exception; only the input type and the interpretation change.
- **A choice about meaning:** choosing among these four is a choice about what "belonging to a cluster" should mean: fuzzy, probabilistic, hierarchical, or relational.



Acknowledgements



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